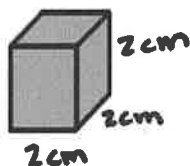
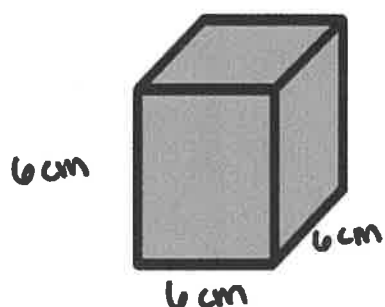


Unit 8: Worked Examples

1. The cube on the left has side lengths of 6cm. The cube on the right has side lengths of 2cm.

- Find the area of one face on each cube.
- Find the volume of each cube.
- Find the surface area of each cube.
- Find out how many smaller cubes will fit inside the larger cube.



Big - 6cm

a. $A = b \cdot h = 6 \cdot 6 = 36 \text{ cm}^2$

b. $V = b \cdot h \cdot w = 6 \cdot 6 \cdot 6 = 216 \text{ cm}^3$

c. $SA = 36 \cdot 6 = 216 \text{ cm}^2$

Small - 2cm

a. $A = b \cdot h = 2 \cdot 2 = 4 \text{ cm}^2$

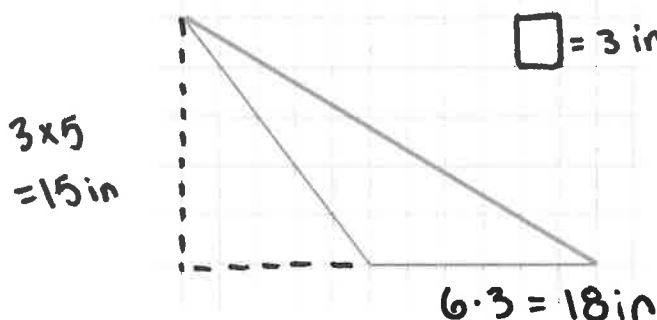
b. $V = b \cdot h \cdot w = 2 \cdot 2 \cdot 2 = 8 \text{ cm}^3$

c. $SA = 4 \cdot 6 = 24 \text{ cm}^2$

d. $\frac{V_{\text{big}}}{V_{\text{little}}} = \frac{216 \text{ cm}^3}{8 \text{ cm}^3}$

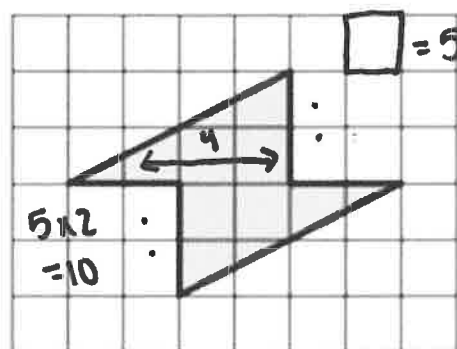
= 27 cubes

2. If each square represents 3 inches what is the area of the shape below?



$$\begin{aligned} \text{Area of Triangle} &= (\text{base} \cdot \text{height}) \cdot \frac{1}{2} \\ &= (18 \cdot 15) \left(\frac{1}{2}\right) \\ &= \frac{270}{1} \cdot \frac{1}{2} \\ &= \frac{270}{2} \\ &= 135 \text{ in}^2 \end{aligned}$$

3. If each square represents 5cm what is the area of the shape below?



$\square = 5 \text{ centimeters (cm)}$

$2 \cdot 5 = 10 \leftarrow \text{height of each triangle}$

$4 \cdot 5 = 20 \leftarrow \text{base of each triangle}$

Area of each triangle

$$\begin{aligned} \frac{1}{2}(b \cdot h) &= \frac{1}{2}(10 \cdot 20) \\ &= \frac{1}{2}(200) \\ &= 100 \end{aligned}$$

Two triangle with the same area means, total area $100 + 100 = 200 \text{ cm}^2$